Reliable Machine Learning with Unreliable Black Boxes **Spring 2026**

Lecture 13: Distribution Shift, Continued

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Overview

In this lecture, we discussed reweighting methods for causal inference, distribution shift, and robustness. The main setting is that we have samples from a source distribution P, but we want to estimate or optimize a quantity under a target distribution Q.

1 Importance sampling

Let P be the source distribution over (X, Y) and let Q be the target distribution. We assume covariate shift:

$$P(Y \mid X) = Q(Y \mid X).$$

We also assume that the support of Q_X is contained in the support of P_X:

$$\mathrm{supp}(Q_X) \subseteq \mathrm{supp}(P_X).$$

Define the importance weight

$$w(x) = \frac{q_X(x)}{p_X(x)}.$$

Then the target risk can be written as a weighted source risk:

$$\begin{aligned} L_Q(h) &= \mathbb{E}_{(X,Y)\sim Q}[\ell(h(X), Y)] \\ &= \mathbb{E}_{(X,Y)\sim P}[w(X)\ell(h(X), Y)]. \end{aligned}$$

Thus, given samples $(x_i, y_i)_{i=1}^n$ from P, we can use weighted empirical risk minimization:

$$\min_{h\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} w(x_i)\ell(h(x_i), y_i).$$

There are two main problems. First, the weights $w(x)$ are usually unknown. Second, even if the weights are correct, the estimator can have large variance.

2 Variance blowup

Fix a hypothesis h and suppose the loss is bounded by B:

$$0 \le \ell(h(x), y) \le B.$$

The weighted estimator is

$$\widehat{L}_Q(h) = \frac{1}{n} \sum_{i=1}^{n} w(x_i) \ell(h(x_i), y_i).$$

Since the samples are independent,

$$\begin{aligned}
\mathrm{Var}[\widehat{L}_Q(h)] &= \frac{1}{n} \mathrm{Var}_{(X,Y)\sim P}[w(X)\ell(h(X), Y)] \\
&\le \frac{1}{n} \mathbb{E}_{(X,Y)\sim P}[w(X)^2 \ell(h(X), Y)^2] \\
&\le \frac{B^2}{n} \mathbb{E}_{X\sim P_X}[w(X)^2].
\end{aligned}$$

Let

$$\mathbb{E}_{X\sim P_X}[w(X)^2] = 1 + \chi^2(Q_X \| P_X).$$

Therefore,

$$\mathrm{Var}[\widehat{L}_Q(h)] \le \frac{B^2}{n} \left(1 + \chi^2(Q_X \| P_X)\right).$$

This means that importance sampling can be unstable when Q_X is far from P_X. A few points with very large weights may dominate the estimator.

3 Raking and calibration

A way to reduce this problem is to avoid estimating the full density ratio. Instead, we only force the reweighted sample to match some known aggregate statistics.

Suppose the original weights are $w_1^{(0)}, \dots, w_n^{(0)}$. Let $\Phi(x)$ be a vector of features, and let t be the target moment vector. The calibration problem is

$$\min_{w_1,\dots,w_n \ge 0} \sum_{i=1}^{n} w_i \log \frac{w_i}{w_i^{(0)}}$$

subject to

$$\sum_{i=1}^{n} w_i = 1, \qquad \sum_{i=1}^{n} w_i \Phi(x_i) = t.$$

The goal is to stay close to the original weights while matching the moments that we trust.

The solution has the form

$$w_i = w_i^{(0)} \exp(\lambda^\top \Phi(x_i) - A(\lambda)),$$

where $A(\lambda)$ normalizes the weights and λ is chosen to satisfy the moment constraint.

This is called raking or calibration of survey weights [Deming and Stephan, 1940, Deville and Särndal, 1992]. In surveys, for example, we may know population totals by age group or region and want the sample weights to match those totals.

4 Contingency tables

A common example is fitting a contingency table. Suppose K_{ij} is an initial table, where rows are regions and columns are age groups. We know the desired row marginals p and column marginals q.

We want a new table Π that is close to K and satisfies

$$\Pi\mathbf{1} = p, \qquad \Pi^\top \mathbf{1} = q.$$

The optimization problem is

$$\min_{\Pi \geq 0} \mathrm{KL}(\Pi \| K)$$

subject to

$$\Pi\mathbf{1} = p, \qquad \Pi^\top \mathbf{1} = q.$$

Here

$$\mathrm{KL}(\Pi \| K) = \sum_{i,j} \left(\Pi_{ij} \log \frac{\Pi_{ij}}{K_{ij}} - \Pi_{ij} + K_{ij} \right).$$

4.1 Lagrangian and KKT conditions

We briefly derive why the solution has the matrix scaling form. The Lagrangian is

$$\mathcal{L}(\Pi, \alpha, \beta) = \mathrm{KL}(\Pi \| K) - \sum_i \alpha_i \left(\sum_j \Pi_{ij} - p_i \right) - \sum_j \beta_j \left(\sum_i \Pi_{ij} - q_j \right).$$

For entries with $\Pi_{ij} > 0$, the stationarity condition gives

$$\frac{\partial \mathcal{L}}{\partial \Pi_{ij}} = \log \frac{\Pi_{ij}}{K_{ij}} - \alpha_i - \beta_j = 0.$$

Therefore,

$$\Pi_{ij} = K_{ij} e^{\alpha_i + \beta_j}.$$

If we define

$$u_i = e^{\alpha_i}, \qquad v_j = e^{\beta_j},$$

then

$$\Pi_{ij} = u_i K_{ij} v_j.$$

Equivalently,

$$\Pi = \mathrm{diag}(u) K \,\mathrm{diag}(v).$$

The remaining KKT conditions enforce the marginal constraints and nonnegativity:

$$\Pi \mathbf{1} = p, \qquad \Pi^\top \mathbf{1} = q, \qquad \Pi_{ij} \geq 0.$$

Thus, the problem reduces to finding scaling vectors u and v so that the row and column sums match the desired marginals.

5 Iterative proportional fitting

The algorithm for finding u and v is iterative proportional fitting, also called Sinkhorn-Knopp scaling [Sinkhorn and Knopp, 1967].

Start with the table K. Then repeat the following two steps:

1. Scale the rows so that the row sums match p.
2. Scale the columns so that the column sums match q.

In terms of scaling vectors, the updates are

$$u \leftarrow p/(Kv), \qquad v \leftarrow q/(K^\top u).$$

6 Optimal transport

Optimal transport is another way to connect a source distribution to a target distribution. Suppose the source has points $x_1, \ldots, x_m$ with masses $p_1, \ldots, p_m$, and the target has points $x'_1, \ldots, x'_n$ with masses $q_1, \ldots, q_n$.

A coupling Π tells us how much mass is moved from source point i to target point j. It must satisfy

$$\sum_j \Pi_{ij} = p_i, \qquad \sum_i \Pi_{ij} = q_j, \qquad \Pi_{ij} \geq 0.$$

If C_{ij} is the cost of moving mass from x_i to x'_j, the optimal transport problem is

$$\min_{\Pi} \sum_{i,j} C_{ij} \Pi_{ij}$$

subject to

$$\sum_j \Pi_{ij} = p_i, \qquad \sum_i \Pi_{ij} = q_j, \qquad \Pi_{ij} \geq 0.$$

After computing Π, one can move source points toward the target distribution. A common transported version of x_i is

$$\widetilde{x}_i = \frac{\sum_j \Pi_{ij} x'_j}{\sum_j \Pi_{ij}}.$$

7 Optimal transport with KL regularization

We also briefly discussed a regularized version of optimal transport. Let K_{ij} be a reference matrix. The goal is to find a coupling Π that satisfies the marginal constraints while staying close to K in KL divergence:

$$\min_{\Pi \geq 0} \mathrm{KL}(\Pi \| K)$$

subject to

$$\sum_j \Pi_{ij} = p_i, \qquad \sum_i \Pi_{ij} = q_j.$$

As in the contingency table problem, the solution has the matrix scaling form

$$\Pi = \mathrm{diag}(u) K \, \mathrm{diag}(v).$$

Thus, the same Sinkhorn scaling algorithm can be used to find the coupling.

This is closely related to entropic optimal transport [Cuturi, 2013].

References

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